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Kuo et al.

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(54) **OPTICAL ELEMENT AND METHOD FOR MANUFACTURING OPTICAL ELEMENT**

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(73) Assignee: **HIMAX TECHNOLOGIES LIMITED**, Tainan (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 302 days.

This patent is subject to a terminal disclaimer.

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(22) Filed: **Nov. 21, 2022**

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US 2024/0045104 A1 Feb. 8, 2024

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(60) Provisional application No. 63/374,003, filed on Aug. 31, 2022.

(51) **Int. Cl.**

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B29D 11/00	(2006.01)
B32B 7/023	(2019.01)
B32B 7/12	(2006.01)
B32B 17/10	(2006.01)
G02B 1/04	(2006.01)
G02B 1/12	(2006.01)

(52) **U.S. Cl.**
CPC ***G02B 1/041*** (2013.01); ***B29D 11/0073*** (2013.01); ***B32B 7/023*** (2019.01); ***B32B 7/12*** (2013.01); ***B32B 17/10*** (2013.01); ***G02B 1/12*** (2013.01); *B32B 2307/412* (2013.01); *B32B 2307/418* (2013.01); *B32B 2551/00* (2013.01); *G02B 6/136* (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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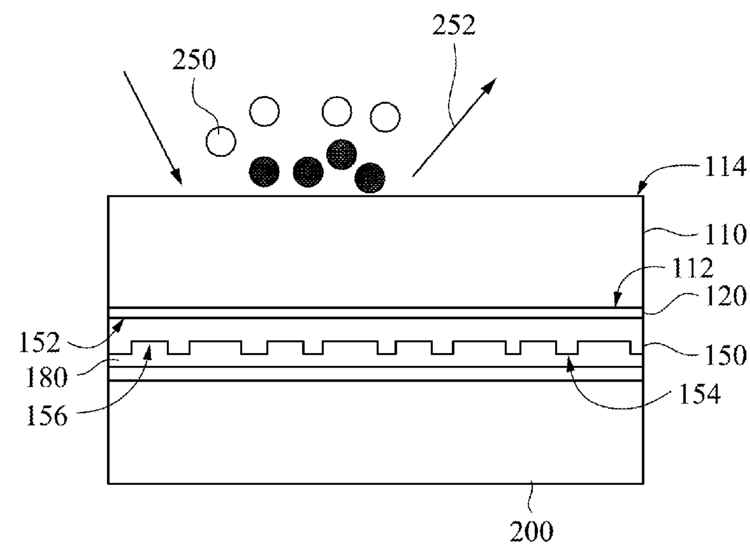
(74) *Attorney, Agent, or Firm* — CKC & Partners Co., LLC

(57) **ABSTRACT**

An optical element and a method for manufacturing the optical element are described. The optical element includes a light penetrating substrate, an optical layer, and an adhesive layer. The optical layer is located on a surface of the light penetrating substrate. The optical layer has a first surface and a second surface, which are opposite to each other. The first surface is set with various diffracting optical structures. A refractive index of the optical layer is ranging from substantially 1 to substantially 4. The adhesive layer is sandwiched between the surface of the light penetrating substrate and the second surface of the optical layer.

20 Claims, 18 Drawing Sheets



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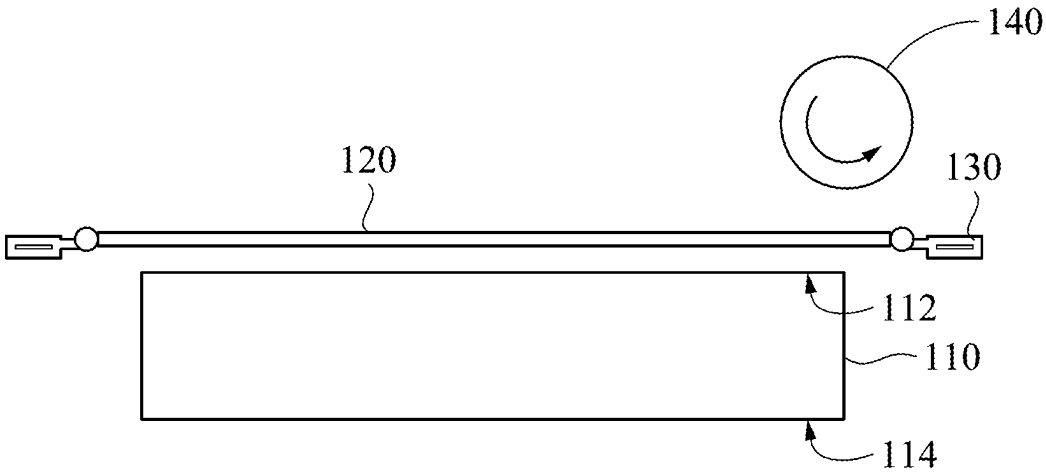

FIG. 1

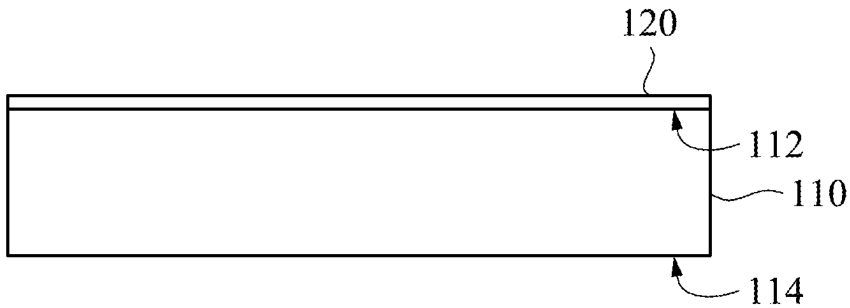

FIG. 2

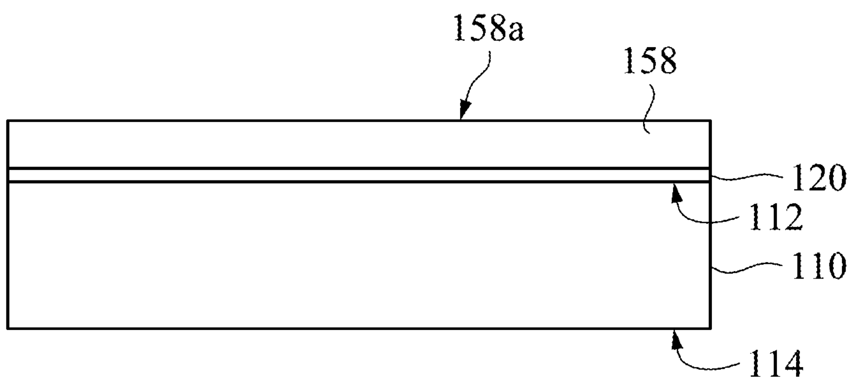

FIG. 3

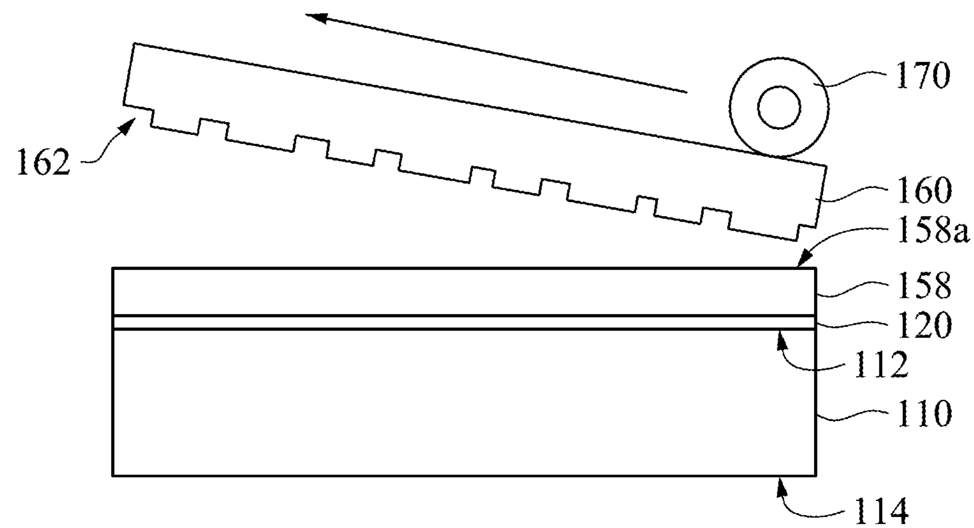
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FIG. 4

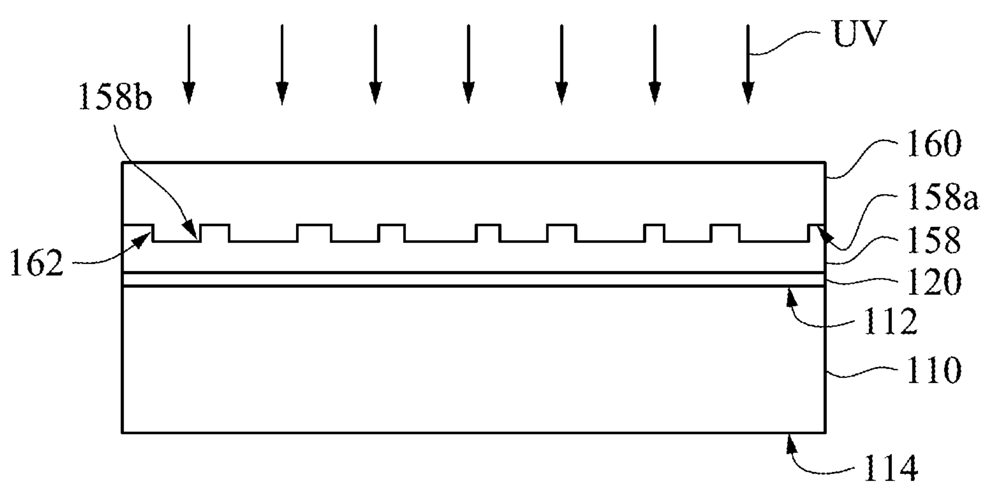
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FIG. 5

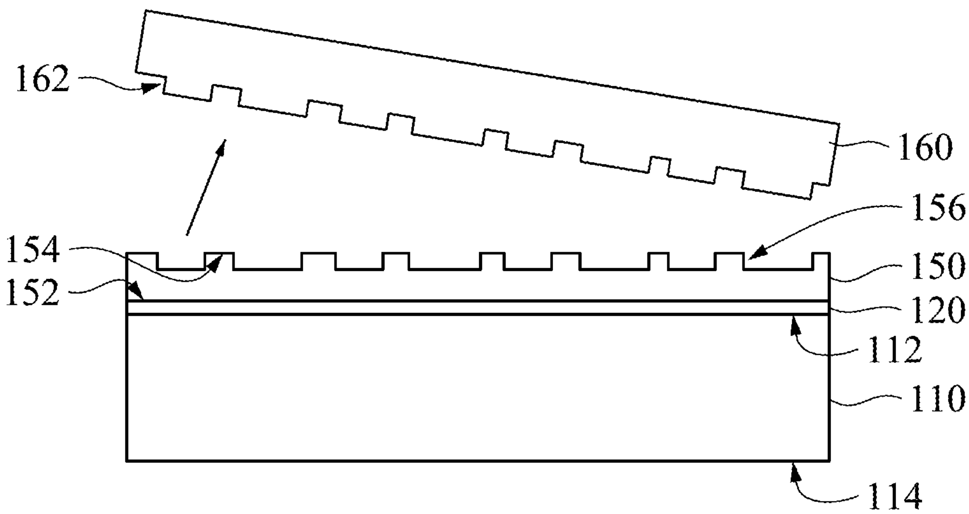
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FIG. 6

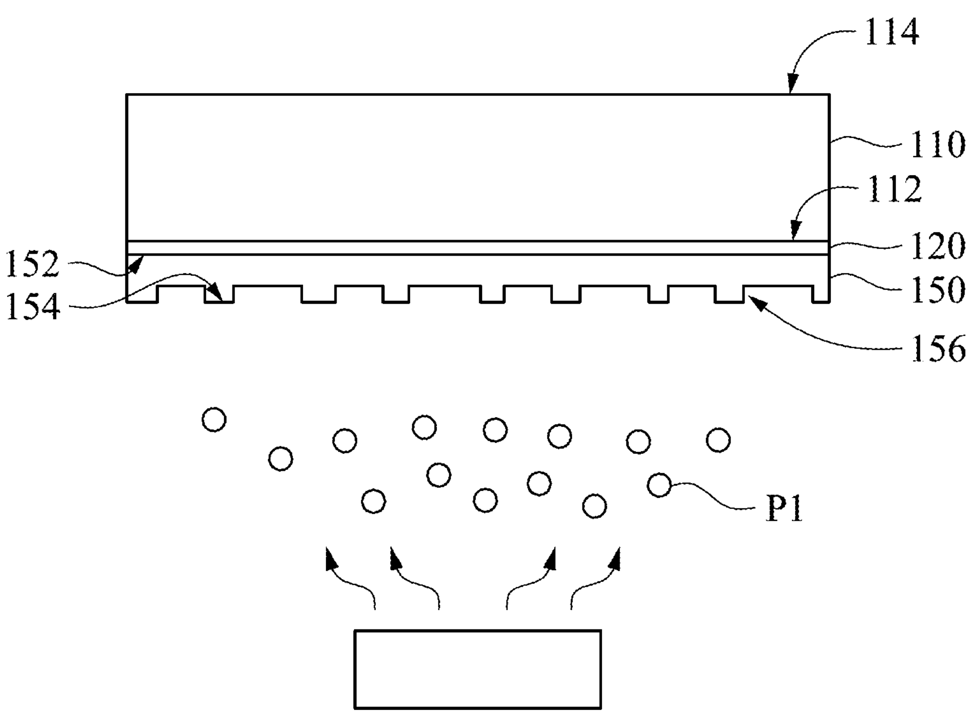
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FIG. 7

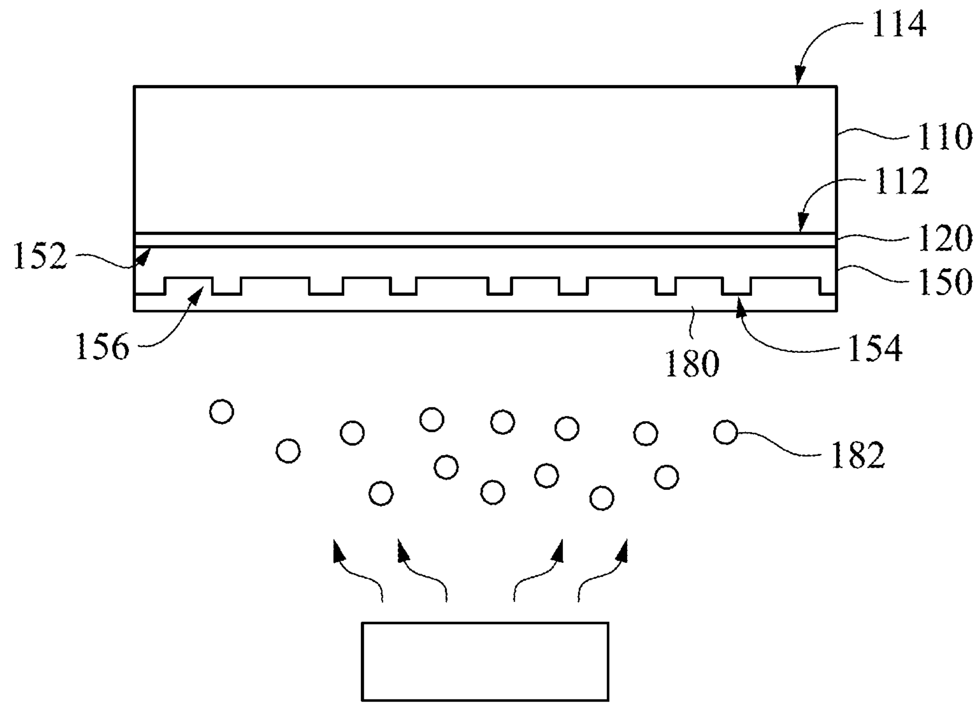
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FIG. 8

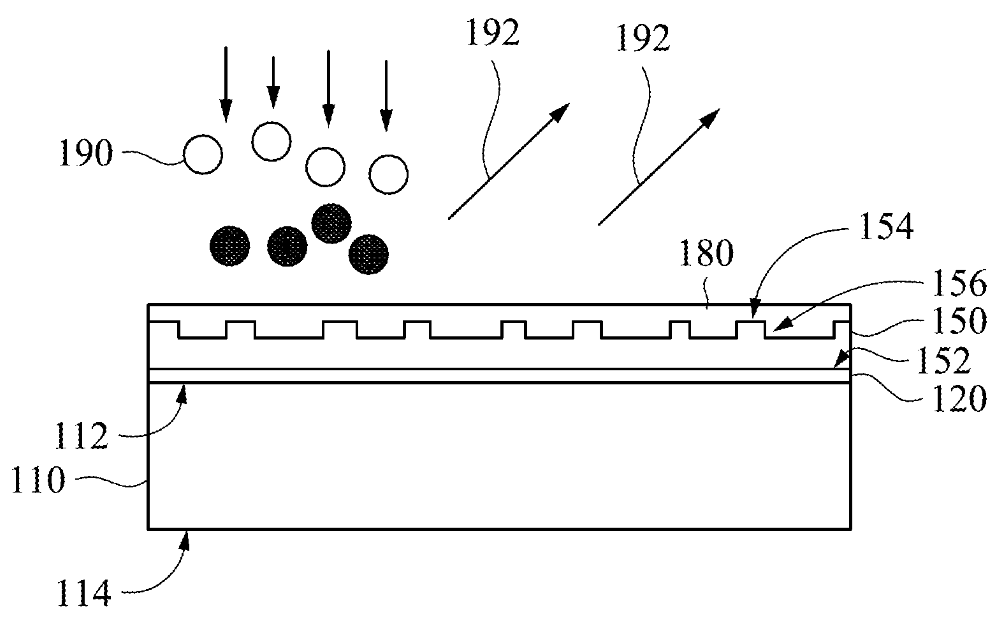
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FIG. 9

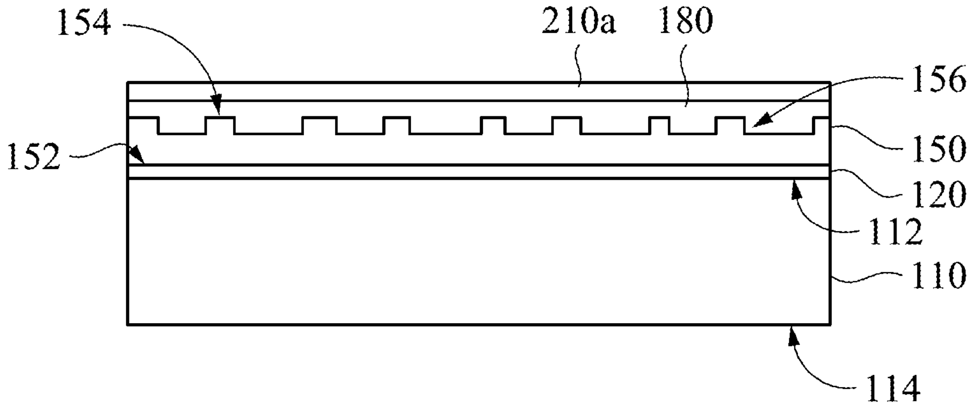

FIG. 10A

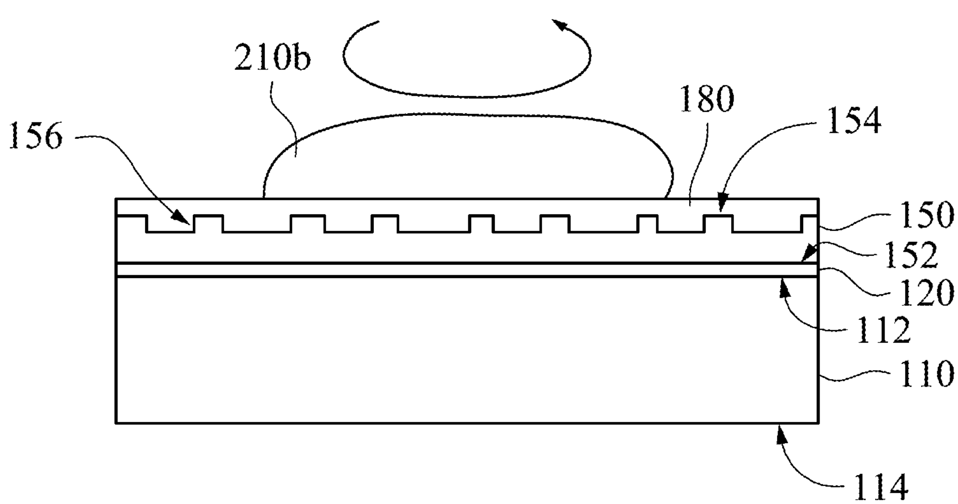

FIG. 10B

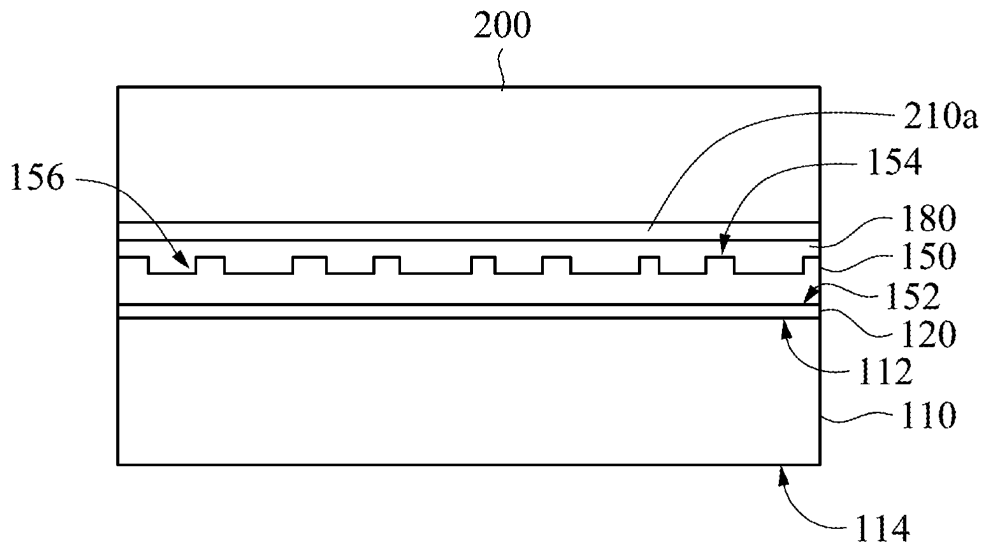

FIG. 11

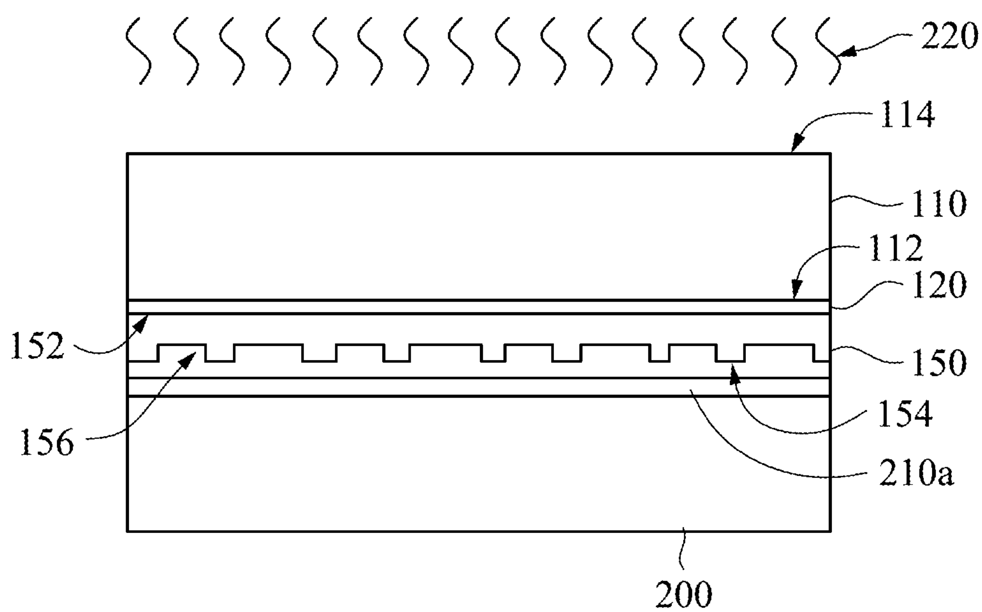

FIG. 12A

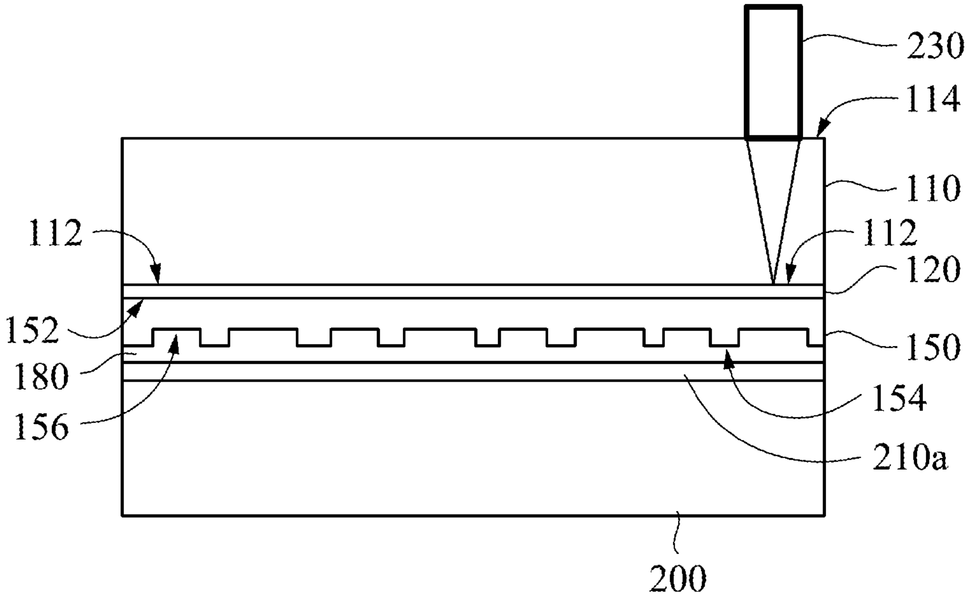

FIG. 12B

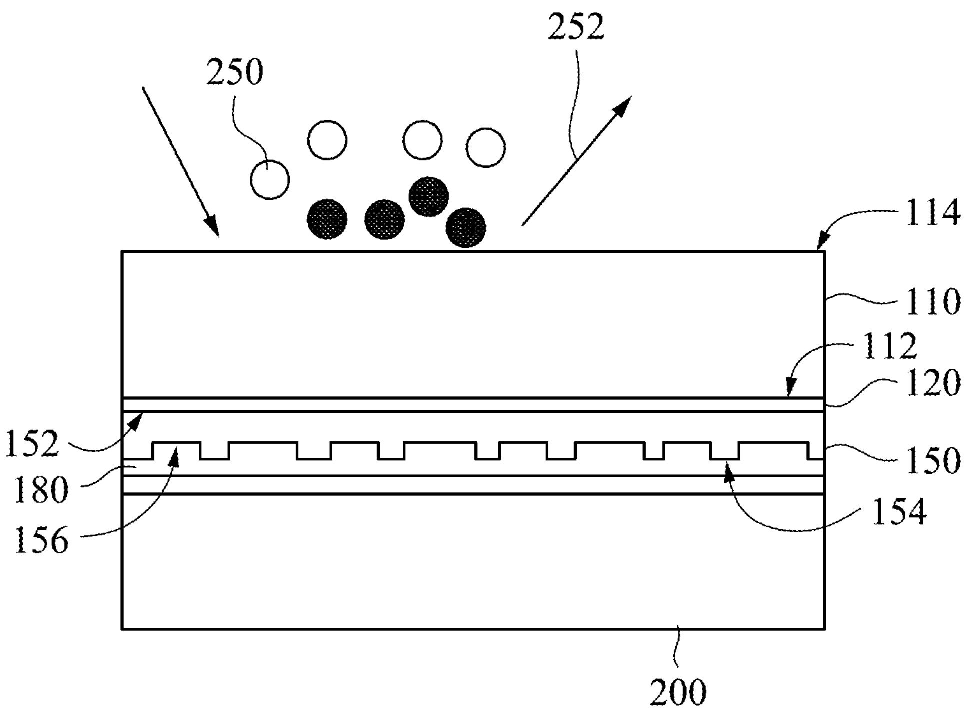

FIG. 12C

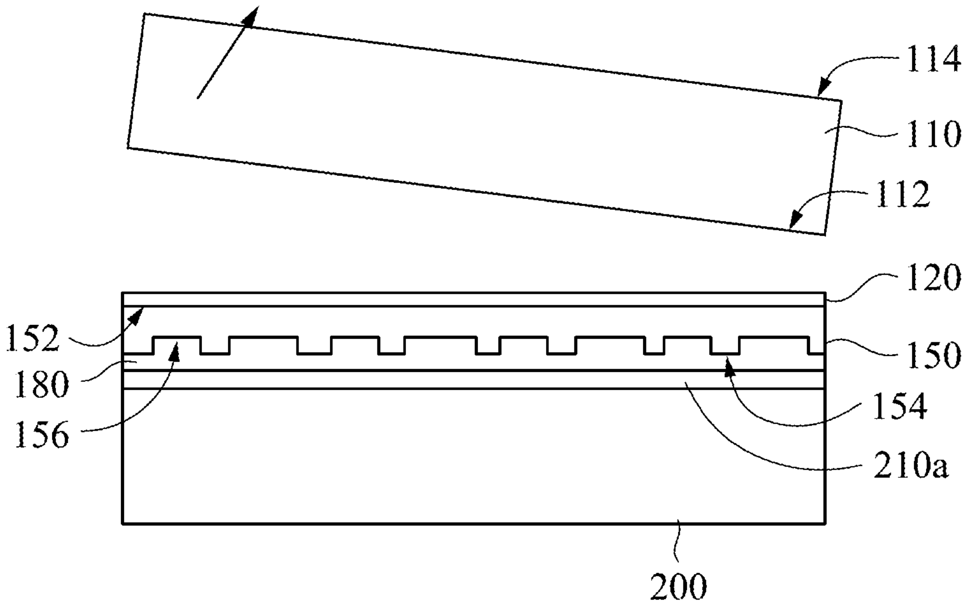
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FIG. 13

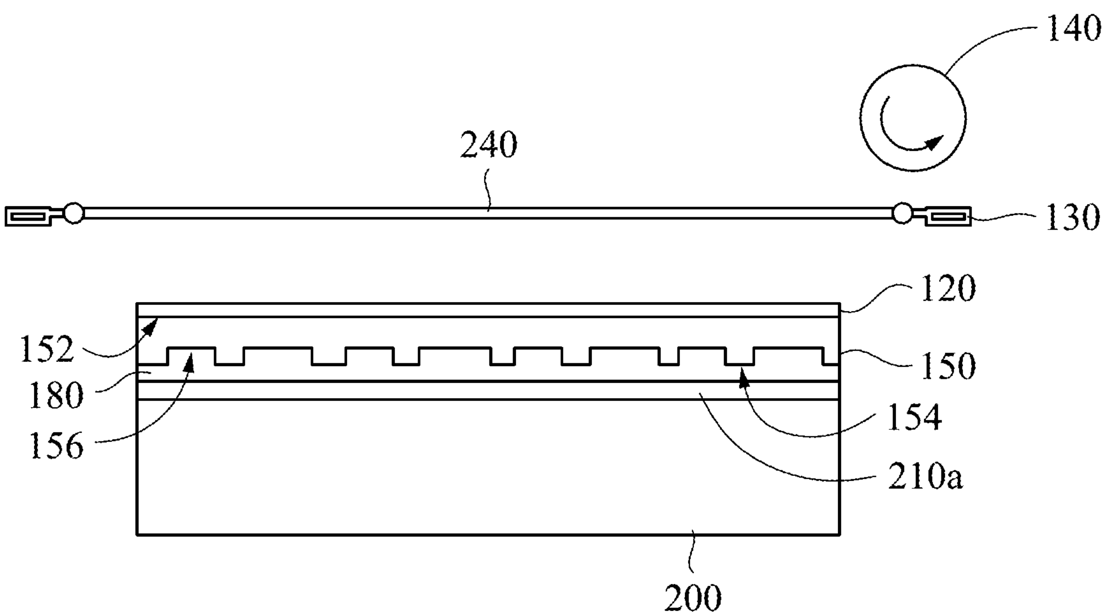
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FIG. 14

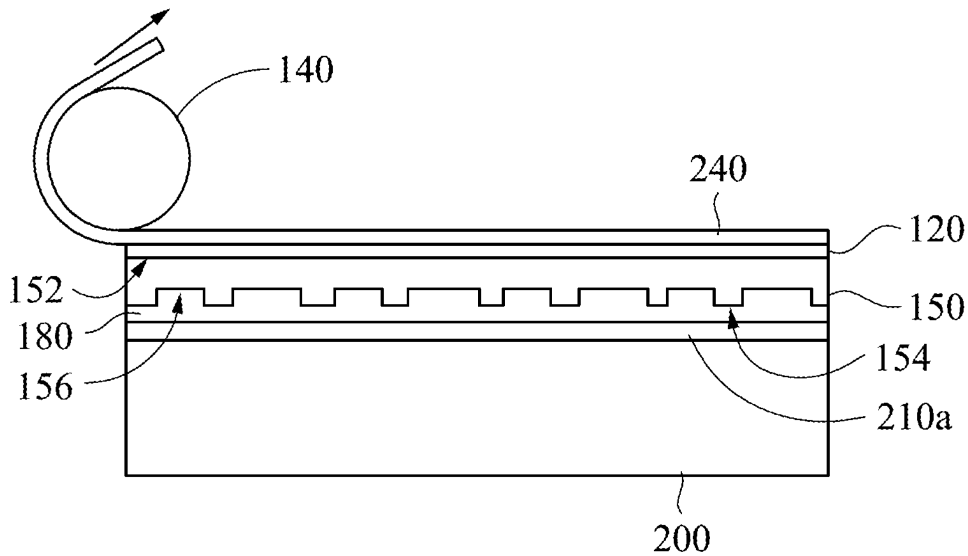

FIG. 15A

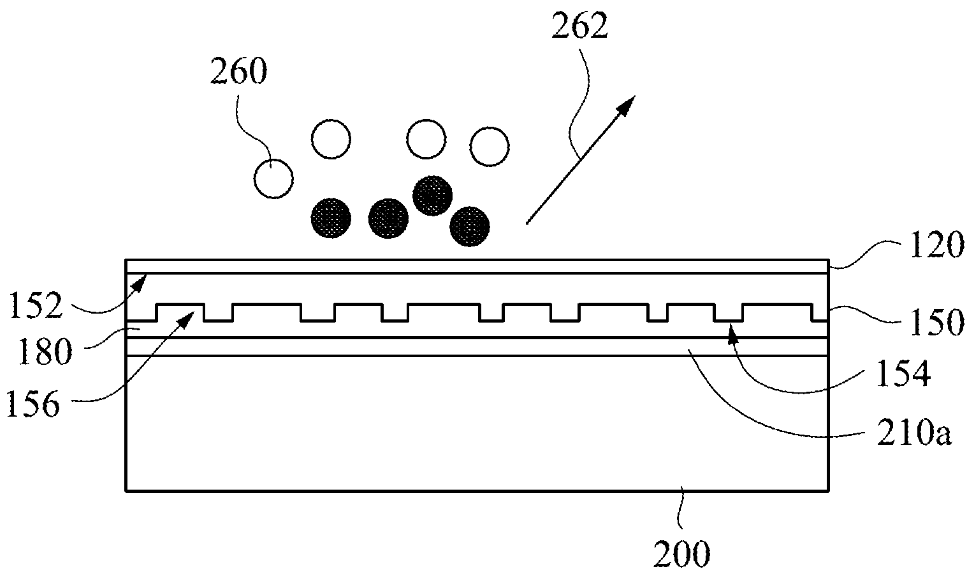

FIG. 15B

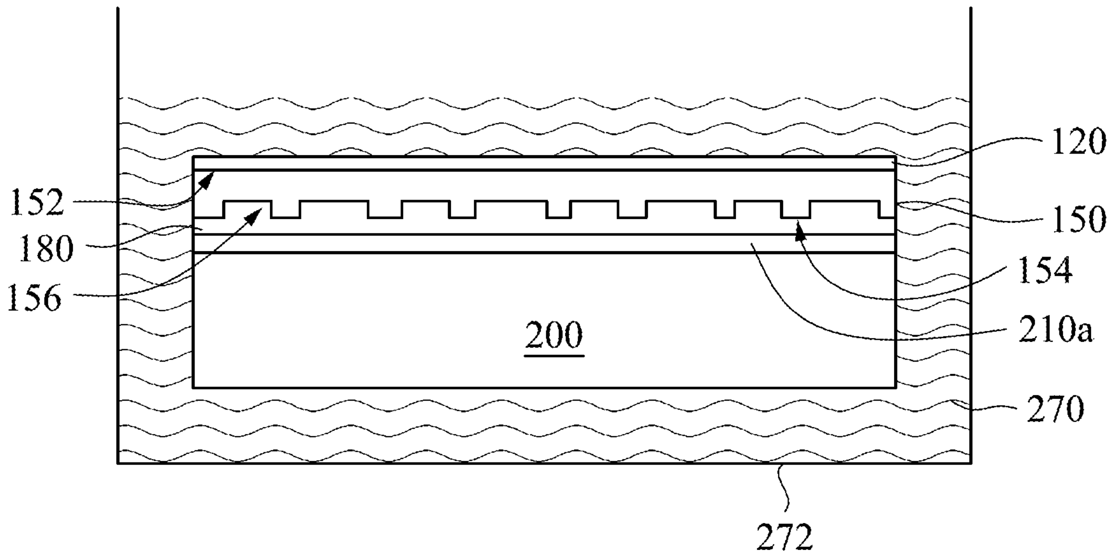

FIG. 15C

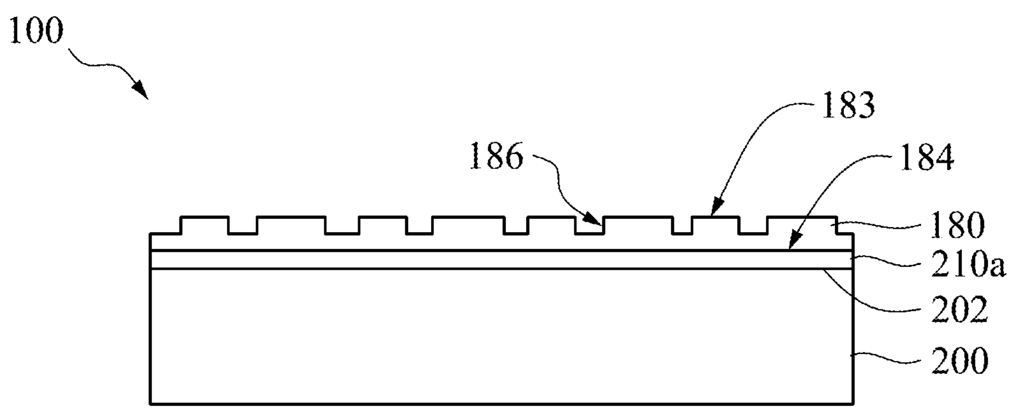

FIG. 16

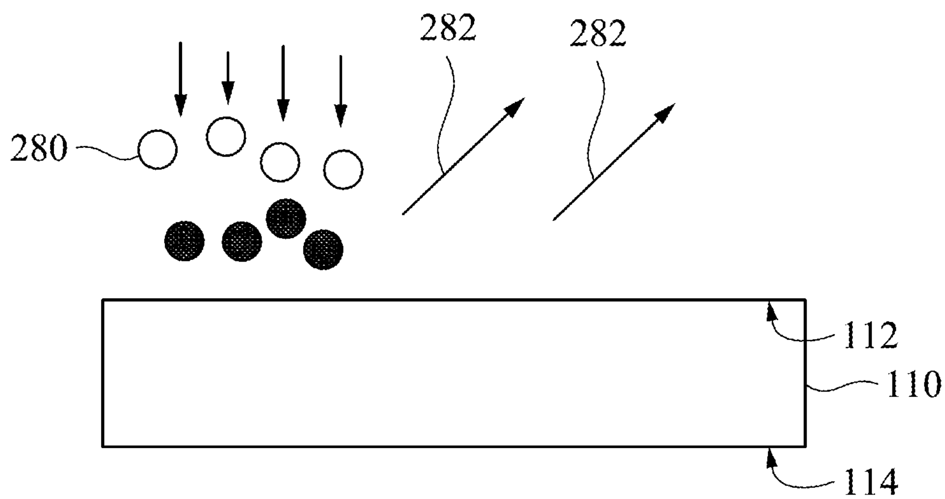

FIG. 17

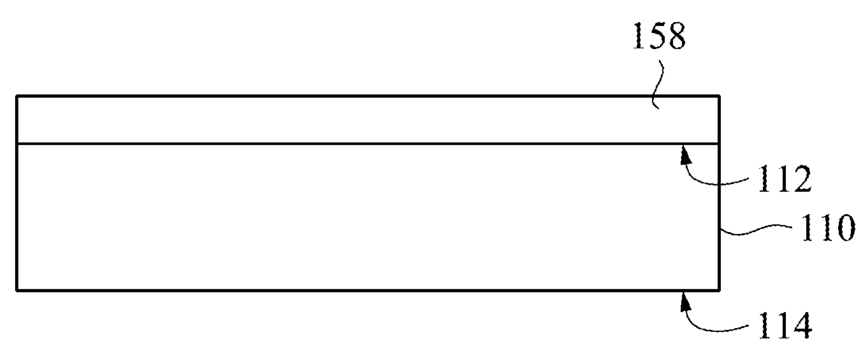

FIG. 18

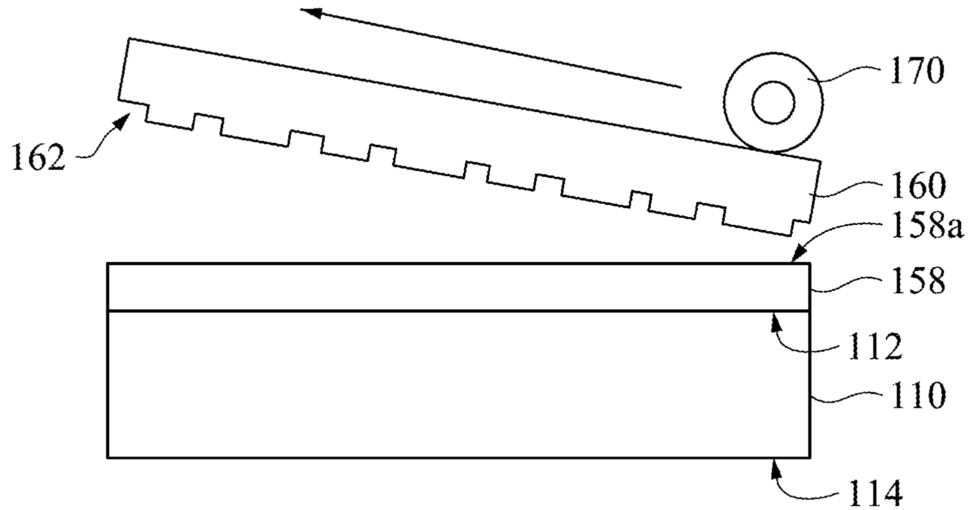
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FIG. 19

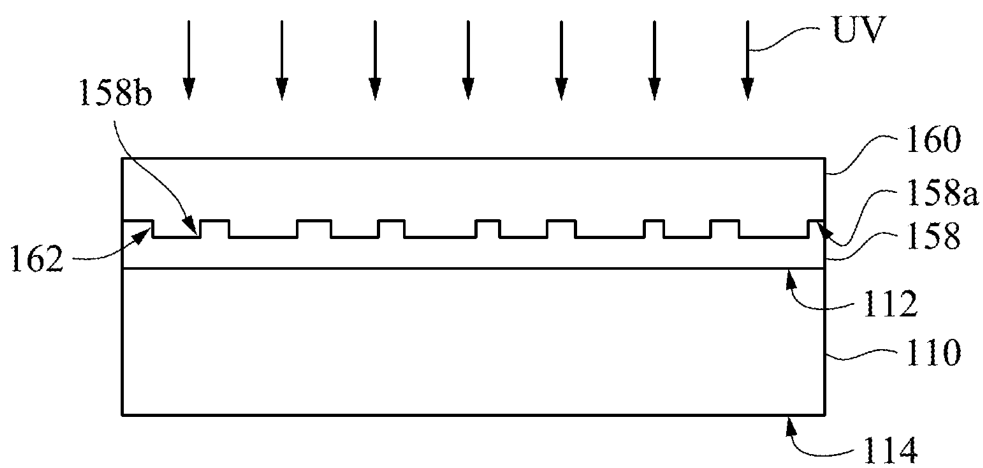
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FIG. 20

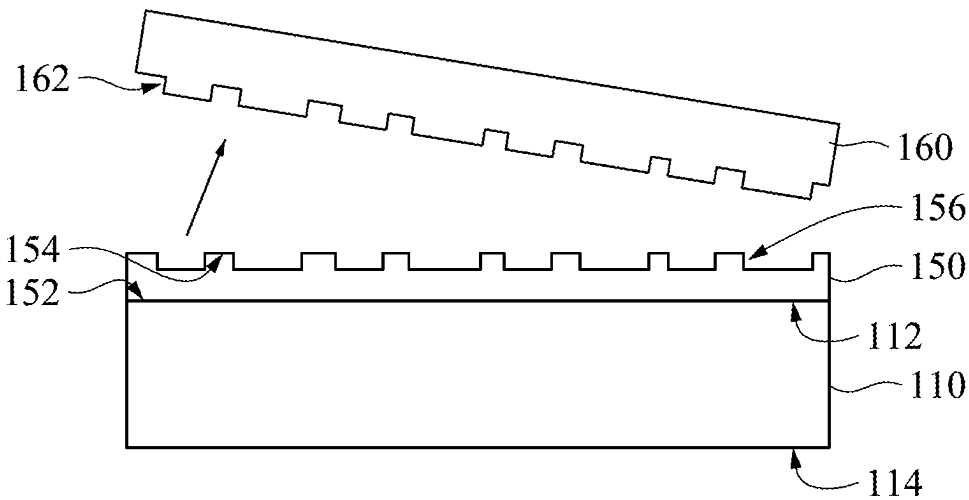
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FIG. 21

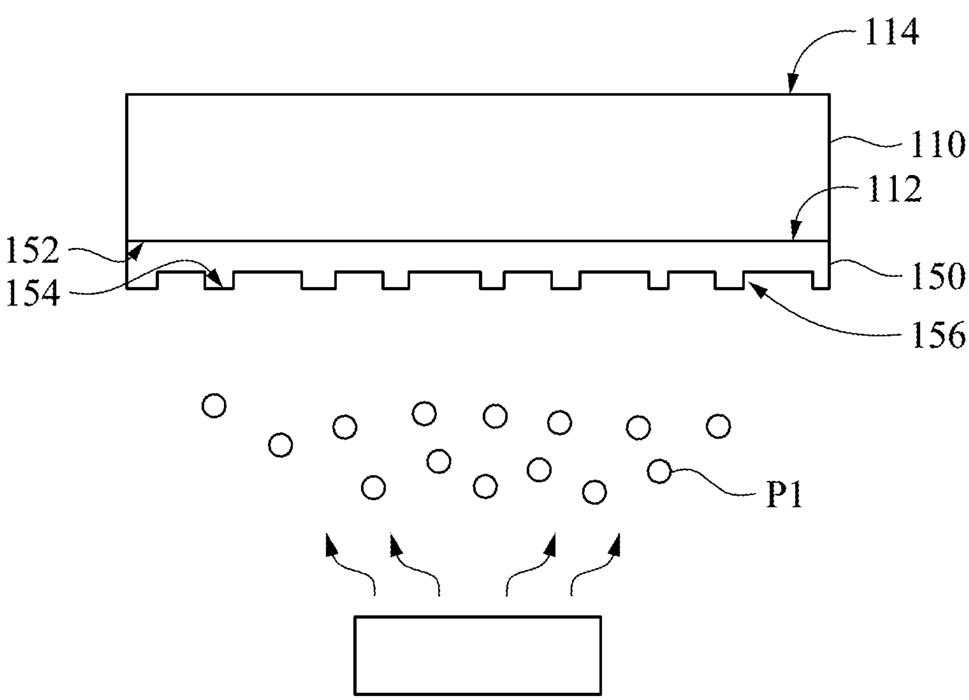
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FIG. 22

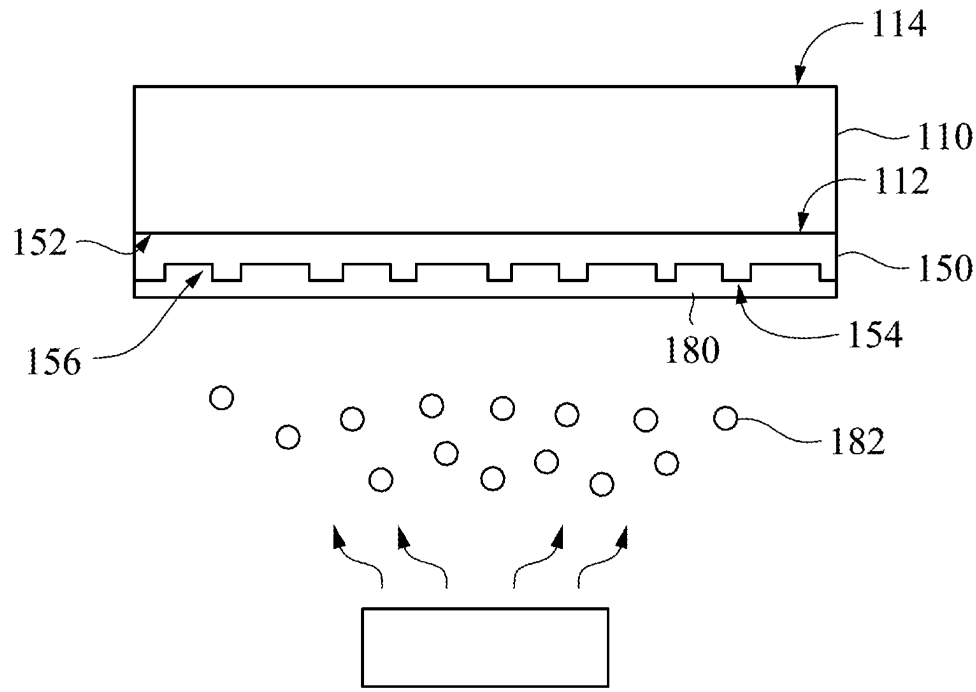
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FIG. 23

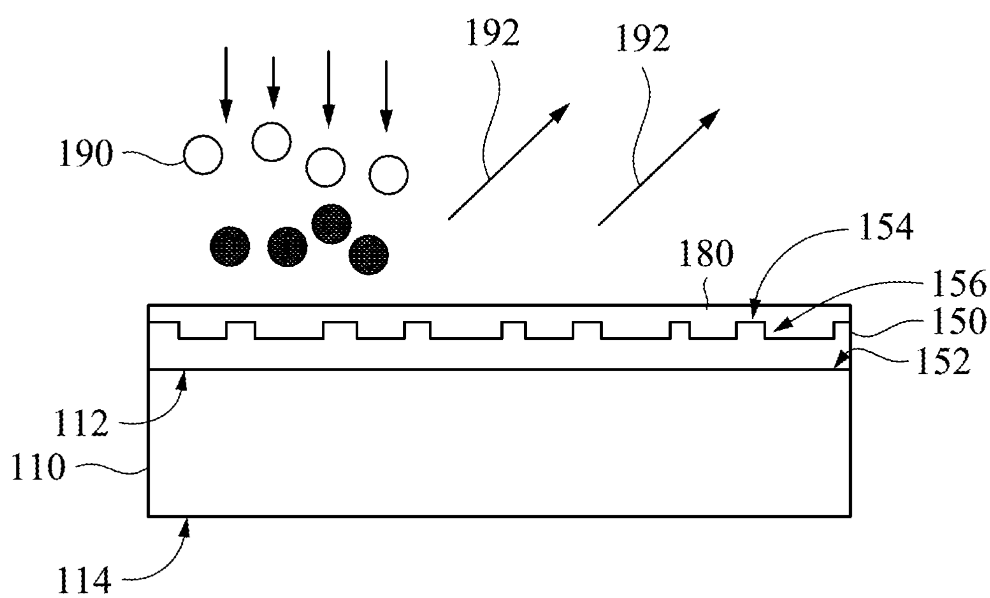
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FIG. 24

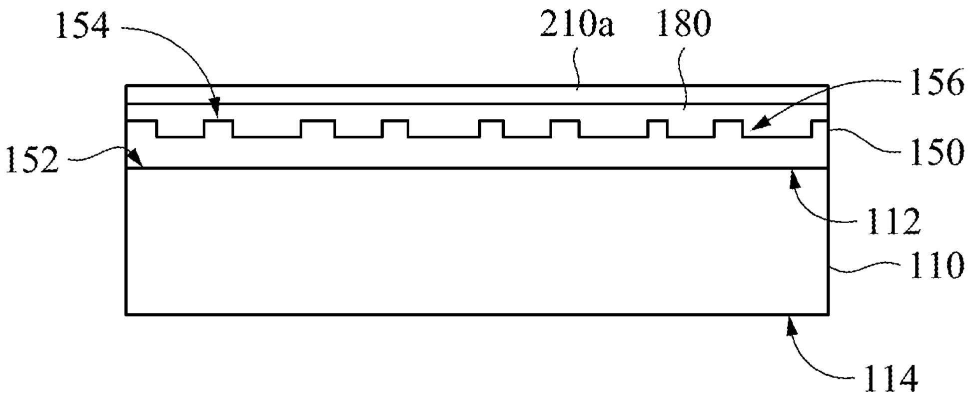

FIG. 25A

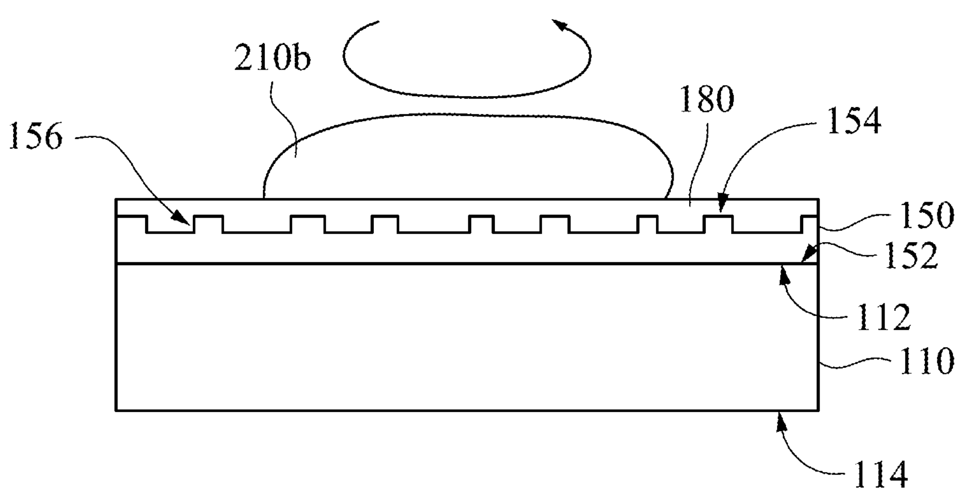

FIG. 25B

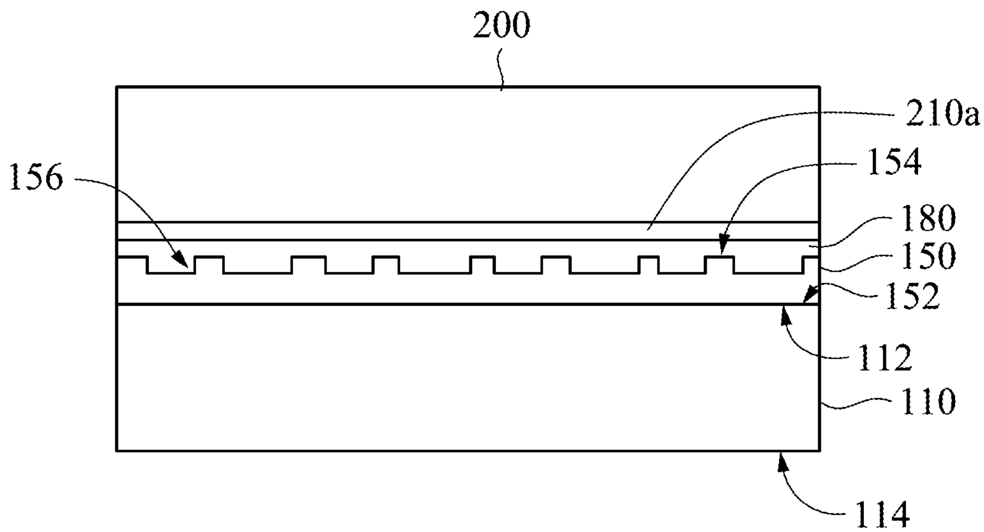

FIG. 26

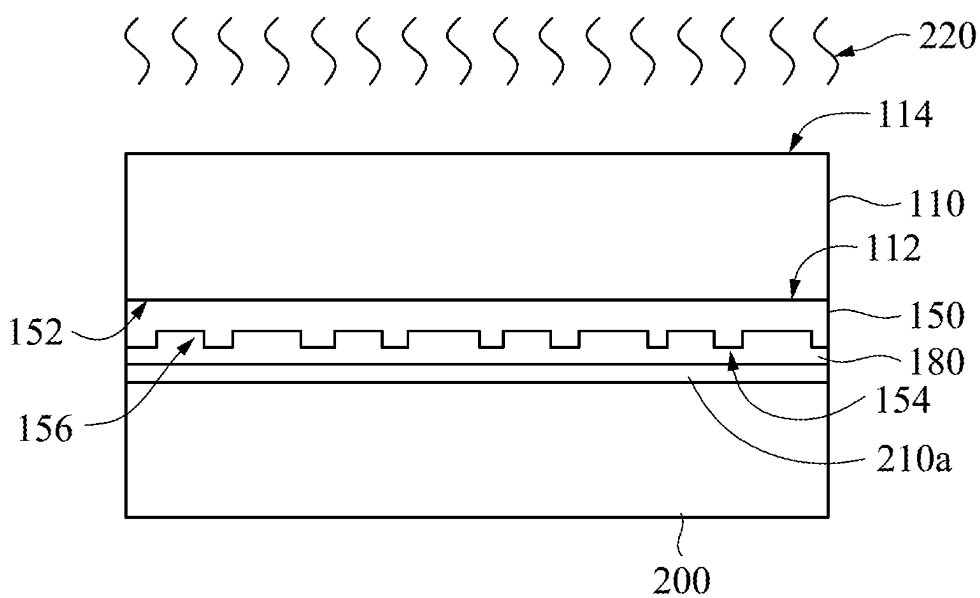

FIG. 27A

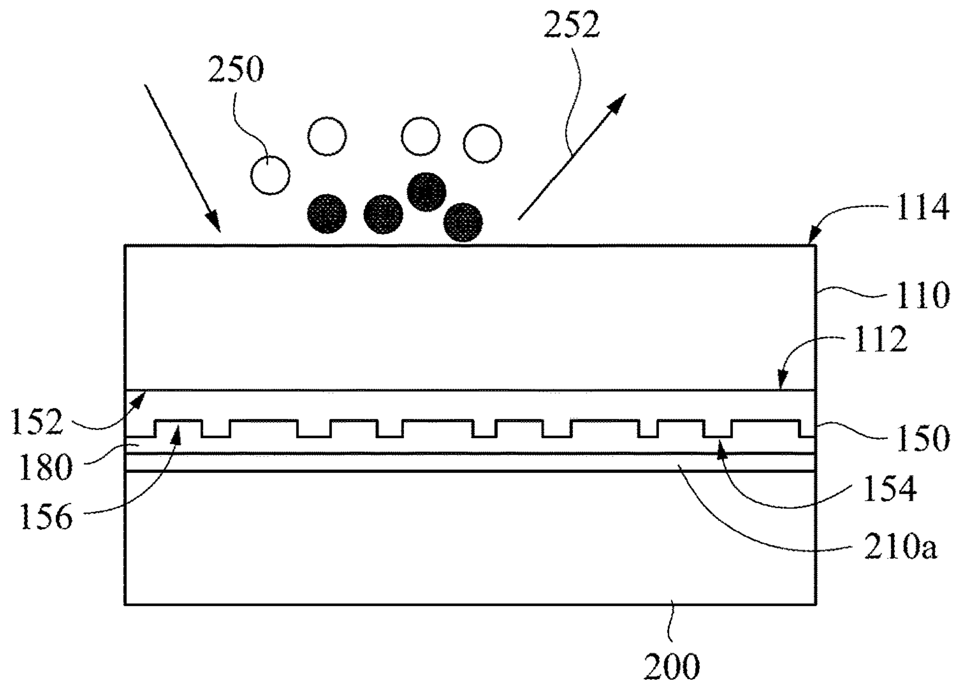

FIG. 27B

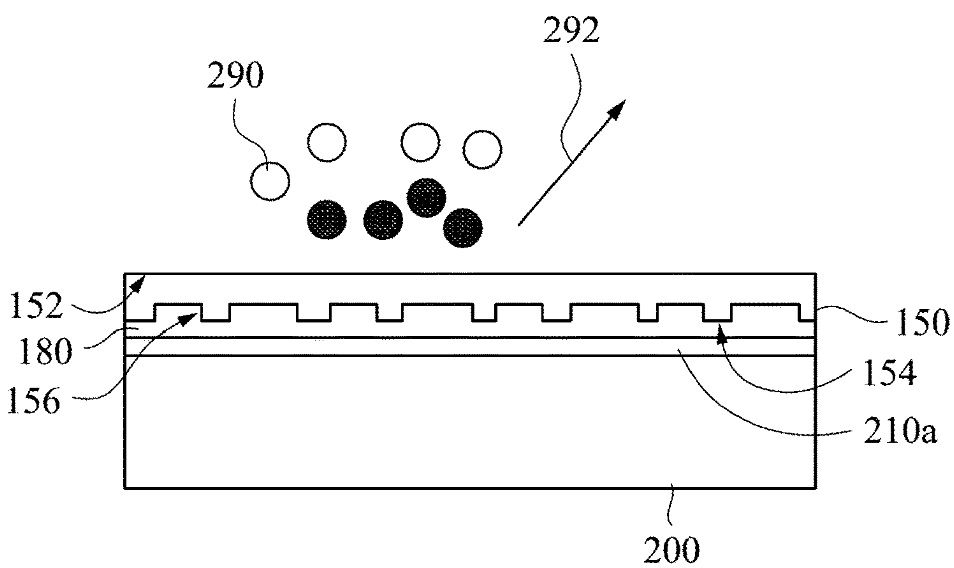

FIG. 28A

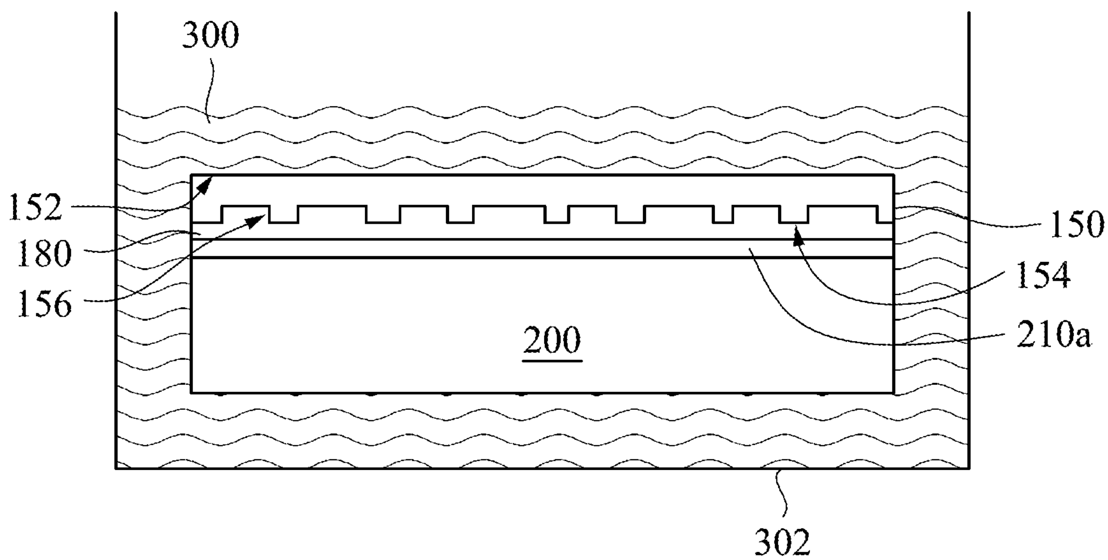

FIG. 28B

OPTICAL ELEMENT AND METHOD FOR MANUFACTURING OPTICAL ELEMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a Continuation-in-part of U.S. application Ser. No. 17/817,346, filed Aug. 3, 2022, now U.S. Pat. No. 11,926,113, issued on Mar. 12, 2024 and also claims priority to U.S. Provisional Application Ser. No. 63/374,003, filed Aug. 31, 2022, all of which are herein incorporated by reference.

BACKGROUND

Field of Invention

The present disclosure relates to an optical technique. More particularly, the present disclosure relates to an optical element and a method for manufacturing the optical element.

Description of Related Art

In the optical field, the using of optical elements with high diffraction angles enables optical devices to have better optical performance. In a convention method for manufacturing an optical element, a glue layer is firstly formed on a light penetrating substrate, and diffracting optical structures are directly imprinted in the glue layer. Thus, the refraction and the diffraction are limited by a refractive index of the glue layer.

SUMMARY

Therefore, one objective of the present disclosure is to provide an optical element and a method for manufacturing the optical element, in which various diffracting optical structures are formed on an optical layer, which is made from a high refractive index material, such that the optical element with a higher diffraction angle is obtained.

According to the above objectives, the present disclosure provides an optical element. The optical element includes a light penetrating substrate, an optical layer, and an adhesive layer. The optical layer is located on a surface of the light penetrating substrate. The optical layer has a first surface and a second surface, which are opposite to each other. The first surface is set with various diffracting optical structures. A refractive index of the optical layer is equal to or greater than 1.4. The adhesive layer is sandwiched between the surface of the light penetrating substrate and the second surface of the optical layer.

According to one embodiment of the present disclosure, the adhesive layer includes an optically clear adhesive.

According to one embodiment of the present disclosure, the adhesive layer includes a pressure sensitive adhesive.

According to one embodiment of the present disclosure, a refractive index of the adhesive layer is ranging from substantially 1 to substantially 4.

According to the above objectives, the present disclosure further provides a method for manufacturing an optical element. In this method, a carrier is provided. A mold layer is formed on the carrier. The mold layer has a first surface and a second surface, which are opposite to each other. The first surface is adjacent to the carrier, and the second surface is set with various microstructures. An anti-sticking treatment is performed on the second surface of the mold layer. An optical layer is formed on the second surface of the mold layer after performing the anti-sticking treatment. The optical layer covers and fills the microstructures. A light penetrating substrate is adhered to the optical layer using an adhesive layer. The optical layer and the light penetrating substrate are respectively located on two opposite sides of the adhesive layer. The carrier is removed from the carrier. The mold layer is removed from the optical layer.

According to one embodiment of the present disclosure, forming the mold layer on the carrier includes coating a glue layer on the carrier, and forming the microstructures on a surface of the glue layer to form the mold layer.

According to one embodiment of the present disclosure, forming the microstructures on the surface of the glue layer includes performing an imprinting step on the surface of the glue layer to press an imprinting mold on the surface of the glue layer, curing the glue layer when the imprinting mold is pressed on the surface of the glue layer, and removing the imprinting mold.

According to one embodiment of the present disclosure, curing the glue layer includes performing an ultraviolet light (UV) exposure treatment, an ultraviolet light exposure with thermal curing treatment, or a thermal curing treatment.

According to one embodiment of the present disclosure, between providing the carrier and forming the mold layer, the method further includes attaching a bonding layer to a surface of the carrier, the mold layer is formed on the bonding layer, and removing the mold layer from the optical layer comprises performing a heating step, a laser ablation step, and/or an etching step on the bonding layer and the mold layer.

According to one embodiment of the present disclosure, between providing the carrier and forming the mold layer, the method further includes attaching a bonding layer to a surface of the carrier, the mold layer is formed on the bonding layer, and removing the mold layer from the optical layer includes performing a chemical soaking step to remove the bonding layer and the mold layer.

According to one embodiment of the present disclosure, performing the anti-sticking treatment includes depositing an anti-sticking material on the second surface of the mold layer, or performing a surface modification treatment on the second surface of the mold layer.

According to one embodiment of the present disclosure, forming the optical layer includes using an atomic layer deposition method, an etching method, a sputtering method, an evaporation method, an imprinting method, or a spin coating method.

According to one embodiment of the present disclosure, a refractive index of the optical layer is ranging from substantially 1 to substantially 4.

According to one embodiment of the present disclosure, between forming the optical layer and adhering the light penetrating substrate to the optical layer, the method further includes performing a plasma cleaning step on the optical layer.

According to one embodiment of the present disclosure, performing the plasma cleaning step includes using an oxygen plasma. The cleaning method is not limited thereto.

According to one embodiment of the present disclosure, adhering the light penetrating substrate to the optical layer using the adhesive layer includes adhering the adhesive layer to the optical layer, and adhering the transparent substrate to the adhesive layer.

According to one embodiment of the present disclosure, the adhesive layer includes a pressure sensitive adhesive.

According to one embodiment of the present disclosure, adhering the light penetrating substrate to the optical layer using the adhesive layer includes coating an optically clear adhesive on the optical layer to form the adhesive layer, and adhering the light penetrating substrate to the adhesive layer.

According to one embodiment of the present disclosure, removing the carrier from the mold layer includes performing a heat treatment to reduce a bonding force between the carrier and the mold layer, and separating the carrier and the mold layer.

According to one embodiment of the present disclosure, removing the carrier from the mold layer includes performing a laser ablation on a bonding layer, in which the bonding layer is disposed between the carrier and the mold layer.

According to one embodiment of the present disclosure, removing the carrier from the mold layer includes performing an etching step on the carrier to reduce the carrier.

According to one embodiment of the present disclosure, removing the mold layer from the optical layer includes performing an etching step on the mold layer.

According to one embodiment of the present disclosure, removing the mold layer from the optical layer includes performing a chemical soaking step to remove the mold layer.

According to one embodiment of the present disclosure, before forming the mold layer on the carrier, the method further includes performing a plasma cleaning step on the carrier.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description in conjunction with the accompanying figures. It is noted that in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, dimensions of the various features can be arbitrarily increased or reduced for clarity of discussion.

FIG. **1** through FIG. **9**, FIG. **10**A, FIG. **11**, FIG. **12**A, FIG. **13**, FIG. **14**, FIG. **15**A, and FIG. **16** are schematic diagrams of intermediate stages in a method for manufacturing of an optical element in accordance with one embodiment of the present disclosure.

FIG. **10**B is a schematic diagram showing forming an adhesive layer on an optical layer in accordance with another embodiment of the present disclosure.

FIG. **12**B is a schematic diagram showing performing a laser ablation step on a bonding layer in accordance with another embodiment of the present disclosure.

FIG. **12**C is a schematic diagram showing performing an etching step on a carrier in accordance with still another embodiment of the present disclosure.

FIG. **15**B is a schematic diagram showing performing an etching step on a bonding layer and a mold layer in accordance with another embodiment of the present disclosure.

FIG. **15**C is a schematic diagram showing performing a chemical soaking step on a bonding layer and a mold layer in accordance with still another embodiment of the present disclosure.

FIG. **17** through FIG. **24**, FIG. **25**A, FIG. **26**, FIG. **27**A, and FIG. **28**A are schematic diagrams of intermediate stages in a method for manufacturing of an optical element in accordance with one embodiment of the present disclosure.

FIG. **25**B is a schematic diagram showing forming an adhesive layer on an optical layer in accordance with another embodiment of the present disclosure.

FIG. **27**B is a schematic diagram showing performing an etching step on a carrier in accordance with another embodiment of the present disclosure.

FIG. **28**B is a schematic diagram showing performing a chemical soaking step on a bonding layer and a mold layer in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

The embodiments of the present disclosure are discussed in detail below. However, it will be appreciated that the embodiments provide many applicable concepts that can be implemented in various specific contents. The embodiments discussed and disclosed are for illustrative purposes only and are not intended to limit the scope of the present disclosure. All of the embodiments of the present disclosure disclose various different features, and these features may be implemented separately or in combination as desired.

In addition, the terms "first", "second", and the like, as used herein, are not intended to mean a sequence or order, and are merely used to distinguish elements or operations described in the same technical terms.

The spatial relationship between two elements described in the present disclosure applies not only to the orientation depicted in the drawings, but also to the orientations not represented by the drawings, such as the orientation of the inversion. Moreover, the terms "connected", "electrically connected", or the like between two components referred to in the present disclosure are not limited to the direct connection or electrical connection of the two components, and may also include indirect connection or electrical connection as required.

Referring to FIG. **1** through FIG. **9**, FIG. **10**A, FIG. **11**, FIG. **12**A, FIG. **13**, FIG. **14**, FIG. **15**A, and FIG. **16**, which are schematic diagrams of intermediate stages in a method for manufacturing of an optical element in accordance with one embodiment of the present disclosure. In the manufacturing of an optical element **100** shown in FIG. **16**, a carrier **110** may be firstly provided, as shown in FIG. **1**. The carrier **110** may be a flat plate. For example, the carrier **110** has two opposite surfaces **112** and **114**, in which at least the surface **112** is a flat surface. The carrier **110** may be a glass flat plate. For example, a thickness of the carrier **110** may be 300 μm.

Next, a bonding layer **120** may be adhered to the surface **112** of the carrier **110**, as shown in FIG. **2**. The surface **112** of the carrier **110** is a flat surface, such that the bonding layer **120** can be smoothly adhered to the surface **112**. Referring to FIG. **1** again, in some examples, the bonding layer **120** is held by a clamping apparatus **130**. Then, the bonding layer **120** may be pressed onto the surface **112** of the carrier **110** by using, for example, a roller **140**. The bonding layer **120** may be an adhesive tape.

After the bonding layer **120** is adhered to the carrier **110**, a mold layer **150** shown in FIG. **6** may be formed on the bonding layer **120**. The mold layer **150** has a first surface **152** and a second surface **154**, in which the first surface **152** and the second surface **154** are opposite to each other. The first surface **152** is adjacent to the bonding layer **120**. For example, the first surface **152** may directly contact with the bonding layer **120**. The second surface **154** is set with microstructures **156**.

In some examples, in the formation of the mold layer **150**, a glue layer **158** is coated on the bonding layer **120**, as shown in FIG. **3**. For example, the glue layer **158** may be formed on the bonding layer **120** by using a spin coating method. Then, the microstructures **156** are formed on a surface **158***a* of the glue layer **158**, such that the mold layer **150** with the microstructures **156** is formed. That is the mold layer **150** is composed of the glue layer **158**. In some examples, in the forming of the microstructures **156**, an imprinting step is performed on the surface **158***a* of the glue layer **158** by using an imprinting mold **160**. The imprinting mold **160** includes a pattern structure **162**. In the imprinting step, as shown in FIG. **4** and FIG. **5**, the imprinting mold **160** is pressed on the surface **158***a* of the glue layer **158** while the glue layer **158** has not been hardened, such that a portion of the glue layer **158** is embedded in the pattern structure **162**. The imprinting step may be performed by using a roller **170** to press the imprinting mold **160** onto the surface **158***a* of the glue layer **158**.

In some examples, as shown in FIG. **5**, when the imprinting mold **160** is pressed on the surface **158***a* of the glue layer **158**, the glue layer **158** is cured to maintain a shape of the surface **158***a* of the glue layer **158**. Thus, after curing, a pattern structure **158***b*, which is opposite to the pattern structure **162** of the imprinting mold **160**, is formed on the surface **158***a* of the glue layer **158**. In some exemplary examples, the glue layer **158** is cured by using an ultraviolet light UV to perform an ultraviolet light exposure treatment on the glue layer **158**. In the example that the glue layer **158** is cured by using the ultraviolet light UV, the imprinting mold **160** is transparent to the ultraviolet light UV. Alternatively, the glue layer **158** is cured by using an ultraviolet light UV to perform with a thermal curing treatment on the glue layer **158**. The glue layer **158** may be cured by performing a thermal curing treatment on the glue layer **158**. A material of the imprinting mold **160** may be, for example, resin polymer, metal, or oxide, but the material of the imprinting mold **160** is not limited thereto. Then, as shown in FIG. **6**, the imprinting mold **160** is removed to complete the formation the microstructures **156** of the mold layer **150**.

After the mold layer **150** is formed, an anti-sticking treatment may be performed on the second surface **154** of the mold layer **150**. In some examples, as shown in FIG. **7**, in the anti-sticking treatment, the stacked structure including the carrier **110**, the bonding layer **120**, and the mold layer **150** is flipped, and an anti-sticking material P1 is deposited on the second surface **154** of the mold layer **150** by using, for example, an evaporation method. In another example, in the anti-sticking treatment, a surface modification treatment is performed on the second surface **154** of the mold layer **150** to make the second surface **154** have an anti-sticking property. The surface modification treatment may be performed by using plasma.

After performing the anti-sticking treatment, the optical layer **180** may be formed on the second surface **154** of the mold layer **150** by using, for example, an atomic layer deposition method, a sputtering method, an evaporation method, an imprinting method, or a spin coating method. The optical layer **180** covers the microstructures **156** of the mold layer **150** and fills the microstructures **156**, such that a surface structure, which is opposite to a topographical structure of the second surface **154** of the mold layer **150**, is formed on the optical layer **180**. In some exemplary examples, as shown in FIG. **8**, an optical material **182** is deposited on the second surface **154** of the mold layer **150** to form the optical layer **180** on the second surface **154**. The optical layer **180** is formed from a high refractive index material. For example, a refractive index of the optical layer **180** may be ranging from substantially 1 to substantially 4.

In some examples, after the optical layer **180** is formed, the stacked structure including the carrier **110**, the bonding layer **120**, the mold layer **150**, and the optical layer **180** is flipped. Then, as shown in FIG. **9**, a plasma cleaning step may be optionally performed on the optical layer **180** to use a plasma **190** to clean the optical layer **180**. For example, the plasma **190** may be an oxygen plasma. During the plasma cleaning step, products **192** are pumped out. The products **192** may be H_2O and/or CO_2.

Next, as shown in FIG. **11**, a light penetrating substrate **200** may be adhered to the optical layer **180** by using an adhesive layer **210***a*, such that the optical layer **180** and the light penetrating substrate **200** are respectively located on two opposite sides of the adhesive layer **210***a*. The light penetrating substrate **200** is a substrate with a high refractive index. In some exemplary examples, the light penetrating substrate **200** may be a glass substrate or a fused silica substrate.

In some examples, as shown in FIG. **10**A, in the operation of adhering the light penetrating substrate **200** to the optical layer **180**, the adhesive layer **210***a* is adhered to the optical layer **180**, and then the light penetrating substrate **200** is placed on and is adhered to the adhesive layer **210***a*. In such examples, the adhesive layer **210***a* is a double-sided tape. The adhesive layer **210***a* may include a pressure sensitive adhesive. The adhesive layer **210***a* is transparent. A haze of the adhesive layer **210***a* may be, for example, smaller than 0.5%, but the present embodiment is not limited thereto. In some exemplary examples, a material of the adhesive layer **210***a* is a high refractive index material. For example, a refractive index of the adhesive layer **210***a* may be ranging from substantially 1 to substantially 4.

In some examples, as shown in FIG. **10**B, in the operation of adhering the light penetrating substrate **200** to the optical layer **180**, an optically clear adhesive **210***b* is coated on the optical layer **180** to form the adhesive layer. The optically clear adhesive **210***b* may be formed on the optical layer **180** by a spin coating method. Next, the light penetrating substrate **200** is placed on and is adhered to the adhesive layer composed of the optically clear adhesive **210***b*. A haze of the optically clear adhesive **210***b* may be, for example, smaller than 0.5%, but the present embodiment is not limited thereto. In some exemplary examples, a material of the optically clear adhesive **210***b* is a high refractive index material. For example, a refractive index of the optically clear adhesive **210***b* may be ranging from substantially 1 to substantially 4.

Then, as shown in FIG. **13**, the carrier **110** is removed from the bonding layer **120**. In some examples, as shown in FIG. **12**A, in the operation of removing the carrier **110**, a heat treatment **220** is performed to reduce a bonding force between the carrier **110** and the bonding layer **120**. Thus, after the heat treatment **220**, the carrier **110** and the bonding layer **120** can be separated from each other easier. In another example, as shown in FIG. **12**B, in the operation of removing the carrier **110**, a laser ablation step is firstly performed on the bonding layer **120** by using a laser device **230**, such that the carrier **110** and the bonding layer **120** can be separated from each other successfully. In still another example, as shown in FIG. **12**C, in the operation of removing the carrier **110**, an etching step is performed on the carrier **110** by using an etchant **250** to reduce the carrier **110** until the carrier **110** is removed. In the examples that the carrier **110** is formed from silicon dioxide (SiO_2), the etchant **250** may be hydrogen fluoride (HF), and products **252** generated during the etching step is silicon fluoride (SiF_4) and water (H_2O). In the examples that the carrier **110** is formed from calcium metasilicate ($CaSiO_3$), the etchant **250** may be hydrogen fluoride, and the products **252** silicon fluoride, water, and calcium fluoride (CaF_2).

After the carrier **110** is removed, the bonding layer **120** and the mold layer **150** may be removed from the optical layer **180**. The bonding layer **120** and the mold layer **150** may be removed simultaneously. For example, in the operation of removing the bonding layer **120** and the mold layer **150**, as shown in FIG. **14**, an adhesive tape **240** may be adhered to the bonding layer **120**. In some examples, the adhesive tape **240** may be held by the clamping apparatus **130**. Then, the adhesive tape **240** may be pressed onto the bonding layer **120** by using the roller **140**. Sequentially, as shown in FIG. **15**A and FIG. **16**, the bonding layer **120** and the mold layer **150** may be pulled by using the adhesive tape **240**, so as to complete the formation of the optical element **100**.

The anti-sticking treatment has been performed on the second surface **154** of the mold layer **150**, such that a bonding force between the second surface **154** of the mold layer **150** and the optical layer **180** is smaller than a bonding surface between the first surface **152** of the mold layer **150** and the bonding layer **120**. Therefore, the mold layer **150** can be separated from the optical layer **180** successfully, and the bonding layer **120** and the mold layer **150** can be easily pulled away.

In another examples, as shown in FIG. **15**B, in the operation of removing the bonding layer **120** and the mold layer **150**, an etching step is performed on the bonding layer **120** and the mold layer **150** by using an etchant **260** to etch away the bonding layer **120** and the mold layer **150**. For example, the etchant **260** may include argon (Ar_2) and chlorine (Cl_2). During the etching step, some products **262** are generated and are pumped out.

In still another examples, as shown in FIG. **15**C, in the operation of removing the bonding layer **120** and the mold layer **150**, a chemical soaking step is performed on the bonding layer **120** and the mold layer **150** by soaking the bonding layer **120** and the mold layer **150** with a chemical **270**. For example, in the chemical soaking step, the stacked structure including light penetrating substrate **200**, the adhesive layer **210***a*, the optical layer **180**, the mold layer **150**, and the bonding layer **120** is immersed in the chemical **270** in a container **272**. After soaking with the chemical **270**, the bonding layer **120** and the mold layer **150** can be removed from the optical layer **180**. In some exemplary examples, the chemical **270** is a residue remover of the EKC series that is researched and developed by the DuPont EKC Technology company.

Referring to FIG. **16** continuously, the optical element **100** includes the light penetrating substrate **200**, the adhesive layer **210***a*, and the optical layer **180**. The optical layer **180** has a first surface **183** and a second surface **184** on two opposite sides of the optical layer **180**. The optical layer **180** is located on a surface **202** of the light penetrating substrate **200**. The adhesive layer **210***a* is sandwiched between the surface **202** of the light penetrating substrate **200** and the second surface **184** of the optical layer **180**. After the mold layer **150** is removed from the first surface **183** of the optical layer **180**, the first surface **183** is formed with diffracting optical structures **186**. For example, each of the diffracting optical structures **186** may be a slanting structure, a binary structure, a stepped structure, a triangle structure, or a trapezoid structure.

Referring to FIG. **17** through FIG. **24**, FIG. **25**A, FIG. **26**, FIG. **27**A, and FIG. **28**A are schematic diagrams of intermediate stages in a method for manufacturing of an optical element in accordance with one embodiment of the present disclosure. In the present embodiment, in the manufacturing of the optical element **100** shown in FIG. **16**, the carrier **110** may be firstly provided. Optionally, before any material layer is formed on the carrier **110**, a plasma cleaning step may be performed on the carrier **110** to use a plasma **280** to clean the carrier **110**. For example, the plasma **280** may be an oxygen plasma. During the plasma cleaning step, products **282** are generated and are pumped out. The products **282** may be H_2O and/or CO_2. In the example that the carrier **110** is clean, the plasma cleaning step can be omitted.

Next, the mold layer **150** shown in FIG. **21** may be directly formed on the surface **112** of the carrier **110**. The first surface **152** and the second surface **154** of the mold layer **150** are opposite to each other, and the first surface **152** is adjacent to the carrier **110**. For example, the first surface **152** may directly contact with the surface **112** of the carrier **110**. The second surface **154** is set with the microstructures **156**.

In some examples, as shown in FIG. **18**, in the formation of the mold layer **150**, the glue layer **158** may be firstly coated on the surface **112** of the carrier **110** by using, for example, a spin coating method. Then, the microstructures **156** are formed on the surface **158***a* of the glue layer **158**, such that the mold layer **150** with the microstructures **156** is formed. In the forming of the microstructures **156**, an imprinting step may be performed on the surface **158***a* of the glue layer **158** by using the imprinting mold **160**. In the imprinting step, as shown in FIG. **19** and FIG. **20**, the imprinting mold **160** is pressed on the surface **158***a* of the glue layer **158** while the glue layer **158** has not been hardened, such that a portion of the glue layer **158** is embedded in the pattern structure **162** of the imprinting mold **160**. The imprinting step may be performed by using the roller **170** to press the imprinting mold **160** onto the surface **158***a* of the glue layer **158**.

In some examples, as shown in FIG. **20**, when the imprinting mold **160** is pressed on the surface **158***a* of the glue layer **158**, the glue layer **158** is cured to maintain a shape of the surface **158***a* of the glue layer **158**. After curing, the pattern structure **158***b*, which is opposite to the pattern structure **162** of the imprinting mold **160**, is formed on the surface **158***a* of the glue layer **158**. For ultraviolet light UV example, the glue layer **158** may be cured to perform an ultraviolet light exposure treatment on the glue layer **158**. The glue layer **158** is cured by using an ultraviolet light UV to perform with a thermal curing treatment on the glue layer **158**. Alternatively, the glue layer **158** may be cured by performing a thermal curing treatment on the glue layer **158**. After curing, the imprinting mold **160** is removed to complete the formation the microstructures **156** of the mold layer **150**, as shown in FIG. **21**.

Then, an anti-sticking treatment may be performed on the second surface **154** of the mold layer **150**. For example, as shown in FIG. **22**, in the anti-sticking treatment, the stacked structure including the carrier **110** and the mold layer **150** may be flipped, and an anti-sticking material P1 may be then deposited on the second surface **154** of the mold layer **150** by using an evaporation method. In another example, in the anti-sticking treatment, a surface modification treatment is performed on the second surface **154** of the mold layer **150** by using, for example, plasma to make the second surface **154** have an anti-sticking property.

After the anti-sticking treatment, the optical layer **180** may be formed on the second surface **154** of the mold layer **150** by using, for example, an atomic layer deposition method, a sputtering method, an evaporation method, an imprinting method, or a spin coating method. As shown in FIG. **23**, the optical layer **180** covers the microstructures **156** of the mold layer **150** and fills concaves of the microstructures **156**, such that a surface structure, which is opposite to a topographical structure of the second surface **154** of the mold layer **150**, is formed on the optical layer **180**. In the forming of the optical layer **180**, the optical material **182** is deposited on the second surface **154** of the mold layer **150**. The optical layer **180** is formed from a high refractive index material. For example, a refractive index of the optical layer **180** may be ranging from substantially 1 to substantially 4.

In some examples, after the optical layer **180** is formed, the stacked structure including the carrier **110**, the mold layer **150**, and the optical layer **180** is flipped. Then, as shown in FIG. **24**, a plasma cleaning step may be optionally performed on the optical layer **180** to clean the optical layer **180**. The plasma cleaning step is performed by using the plasma **190**, such as an oxygen plasma. During the plasma cleaning step, the products **192** are pumped out. The products **192** may be H_2O and/or CO_2.

Subsequently, as shown in FIG. **26**, the light penetrating substrate **200** may be adhered to the optical layer **180** by using the adhesive layer **210***a*. Thus, the optical layer **180** and the light penetrating substrate **200** are respectively located on two opposite sides of the adhesive layer **210***a*.

In some examples, as shown in FIG. **25**A, in the operation of adhering the light penetrating substrate **200** to the optical layer **180**, the adhesive layer **210***a* is firstly adhered to the optical layer **180**, and the light penetrating substrate **200** is placed on and is adhered to the adhesive layer **210***a*. In the examples, the adhesive layer **210***a* is a double-sided tape. The adhesive layer **210***a* may include a pressure sensitive adhesive. A refractive index of the adhesive layer **210***a* may be ranging from substantially 1 to substantially 4.

In some examples, as shown in FIG. **25**B, in the operation of adhering the light penetrating substrate **200** to the optical layer **180**, the optically clear adhesive **210***b* is coated on the optical layer **180** to form the adhesive layer by using, for example, a spin coating method. Then, the light penetrating substrate **200** is placed on and is adhered to the adhesive layer composed of the optically clear adhesive **210***b*. A refractive index of the optically clear adhesive **210***b* may be ranging from substantially 1 to substantially 4.

After the light penetrating substrate **200** is adhered to the optical layer **180**, the carrier **110** is removed from the bonding layer **120**. In some examples, as shown in FIG. **27**A, in the operation of removing the carrier **110**, a heat treatment **220** is performed to reduce a bonding force between the carrier **110** and the mold layer **150**. After the heat treatment **220**, the carrier **110** and the bonding layer **120** can be separated from each other easier.

In another example, as shown in FIG. **27**B, in the operation of removing the carrier **110**, an etching step is performed on the carrier **110** by using the etchant **250** to reduce the carrier **110** until the carrier **110** is removed. In the examples that the carrier **110** is formed from silicon dioxide, the etchant **250** may be hydrogen fluoride, and the products **252** generated during the etching step is silicon fluoride and water. In the examples that the carrier **110** is formed from calcium metasilicate, the etchant **250** may be hydrogen fluoride, and the products **252** silicon fluoride, water, and calcium fluoride.

After the carrier **110** is removed, the mold layer **150** may be removed from the optical layer **180**. In some examples, as shown in FIG. **28**A, in the operation of removing the mold layer **150**, an etching step is performed on the bonding layer **120** and the mold layer **150** by using an etchant **290** to etch away the mold layer **150**. For example, the etchant **290** may include argon and chlorine. During the etching step, some products **292** are generated and are pumped out.

In another examples, as shown in FIG. **28**B, in the operation of removing the mold layer **150**, a chemical soaking step is performed on the mold layer **150** by soaking the mold layer **150** with a chemical **300**. For example, in the chemical soaking step, the stacked structure including light penetrating substrate **200**, the adhesive layer **210***a*, the optical layer **180**, and the mold layer **150** is immersed in the chemical **300** in a container **302**. After soaking with the chemical **300**, the mold layer **150** can be removed from the optical layer **180**, so as to complete the formation of the optical element **100**. In some exemplary examples, the chemical **300** is a residue remover of the EKC series that is researched and developed by the DuPont EKC Technology company.

According to the embodiments described above, one advantage of the present disclosure is that various diffracting optical structures are formed on an optical layer, which is made from a high refractive index material, such that the optical element with a higher diffraction angle is obtained.

The features of several embodiments are outlined above, so those skilled in the art can understand the aspects of the present disclosure. Those skilled in the art will appreciate that the present disclosure can be readily utilized as a basis for designing or modifying other processes and structures, thereby achieving the same objectives and/or achieving the same advantages as the embodiments described herein. Those skilled in the art should also understand that these equivalent constructions do not depart from the spirit and scope of the present disclosure, and they can make various changes, substitutions, and alteration without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method for manufacturing an optical element, and the method comprising:

providing a carrier;

forming a mold layer on the carrier, wherein the mold layer has a first surface and a second surface, which are opposite to each other, the first surface is adjacent to the carrier, and the second surface is set with a plurality of microstructures;

performing an anti-sticking treatment on the second surface of the mold layer;

forming an optical layer on the second surface of the mold layer after performing the anti-sticking treatment, wherein the optical layer covers and fills the microstructures;

adhering a light penetrating substrate to the optical layer using an adhesive layer, wherein the optical layer and the light penetrating substrate are respectively located on two opposite sides of the adhesive layer;

removing the carrier from the mold layer; and

removing the mold layer from the optical layer.

2. The method of claim **1**, wherein forming the mold layer on the carrier comprises:

coating a glue layer on the carrier; and

forming the microstructures on a surface of the glue layer to form the mold layer.

3. The method of claim **2**, wherein forming the microstructures on the surface of the glue layer comprises:

performing an imprinting step on the surface of the glue layer to press an imprinting mold on the surface of the glue layer;

curing the glue layer when the imprinting mold is pressed on the surface of the glue layer; and

removing the imprinting mold.

4. The method of claim **3**, wherein curing the glue layer comprises performing an ultraviolet light exposure treatment, an ultraviolet light exposure with thermal curing treatment, or a thermal curing treatment.

5. The method of claim **1**, wherein between providing the carrier and forming the mold layer, the method further comprises attaching a bonding layer to a surface of the carrier, the mold layer is formed on the bonding layer, and removing the mold layer from the optical layer comprises performing a heating step, a laser ablation step, and/or an etching step on the bonding layer and the mold layer.

6. The method of claim **1**, wherein between providing the carrier and forming the mold layer, the method further comprises attaching a bonding layer to a surface of the carrier, the mold layer is formed on the bonding layer, and removing the mold layer from the optical layer comprises performing a chemical soaking step to remove the bonding layer and the mold layer.

7. The method of claim **1**, wherein performing the anti-sticking treatment comprises depositing an anti-sticking material on the second surface of the mold layer, or performing a surface modification treatment on the second surface of the mold layer.

8. The method of claim **1**, wherein forming the optical layer comprises using an atomic layer deposition method, an etching method, a sputtering method, an evaporation method, an imprinting method, or a spin coating method.

9. The method of claim **1**, wherein a refractive index of the optical layer is ranging from substantially 1 to substantially 4.

10. The method of claim **1**, wherein between forming the optical layer and adhering the light penetrating substrate to the optical layer, the method further comprises performing a plasma cleaning step on the optical layer.

11. The method of claim **10**, wherein performing the plasma cleaning step comprises using an oxygen plasma.

12. The method of claim **1**, wherein adhering the light penetrating substrate to the optical layer using the adhesive layer comprises:

adhering the adhesive layer to the optical layer; and

adhering the light penetrating substrate to the adhesive layer.

13. The method of claim **12**, wherein the adhesive layer comprises a pressure sensitive adhesive.

14. The method of claim **1**, wherein adhering the light penetrating substrate to the optical layer using the adhesive layer comprises:

coating an optically clear adhesive on the optical layer to form the adhesive layer; and

adhering the light penetrating substrate to the adhesive layer.

15. The method of claim **1**, wherein removing the carrier from the mold layer comprises:

performing a heat treatment to reduce a bonding force between the carrier and the mold layer; and

separating the carrier and the mold layer.

16. The method of claim **1**, wherein removing the carrier from the mold layer comprises:

performing a laser ablation on a bonding layer, wherein the bonding layer is disposed between the carrier and the mold layer.

17. The method of claim **1**, wherein removing the carrier from the mold layer comprises:

performing an etching step on the carrier to reduce the carrier.

18. The method of claim **1**, wherein removing the mold layer from the optical layer comprises performing an etching step on the mold layer.

19. The method of claim **1**, wherein removing the mold layer from the optical layer comprises performing a chemical soaking step to remove the mold layer.

20. The method of claim **1**, wherein before forming the mold layer on the carrier, the method further comprises performing a plasma cleaning step on the carrier.

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